

9 Types of API Testing

First, think of API testing as checking whether your backend APIs work correctly, work with other systems, remain reliable under heavy traffic, and are secure.

Let's say, your application has:

POST /api/login

GET /api/users/123

POST /api/appointments

GET /api/appointments

DELETE /api/appointments/123

Now follow 9 types of API testing.

1. Smoke Testing		
2. Functional Testing		
3. Integration Testing		

1. Smoke Testing

Purpose: Quickly check whether the API is basically working after a new deployment or build.

Think like "Is the system alive enough to start deeper testing?"

Example:

After deploying your backend, you might test:

</> http

GET /api/health

Expected:

</> JSON

```
{  
  "status": "healthy"  
}
```

Then test:

</> http

POST /api/login

With valid credentials.

If these basic APIs work, you proceed with more detailed testing.

So, what it checks

- API server is running
- Major endpoints respond
- Database connection is working
- Basic authentication works
- No immediate deployment failure

Real-world example

You deploy a Node.js application to AWS.

First, you check:

Application running?

/api/health works?

Login works/

Database reachable?

If /api/health doesn't work, there's little point running hundreds of detailed tests.

Smoke testing = basic "is it alive?" test.

2. Functional Testing

Purpose: To verify that each API does exactly what it is supposed to do.

Think like: "Does this API actually perform its intended functional?"

Suppose you have:

</> http

POST /api/users

Request:

</> JSON

{

"name": "John",

"email": "john@example.com"

}

Expected:

</> http

201 Created

and:

</> JSON

{

"id": 101,

"name": "John",

"email": "john@example.com"

}

You test things like:

Valid input

</> Json

{

"name": "John",

"email": "john@example.com"

}

Expected:

201 Created

Missing email

</> JSON

{

"name": "John"

}

Expected:

400 Bad Request

Duplicate email

Expected something like:

409 Conflict

Invalid email

</> JSON

{

"name": "John",

"email": "hello"

}

Expected:

400 Bad Request

Here, functional testing verifies the business rules of the API.

3. Integration Testing

Purpose: Check whether multiple components/systems work correctly together.